POWER WINDOW MAIN SWITCH INSPECTION

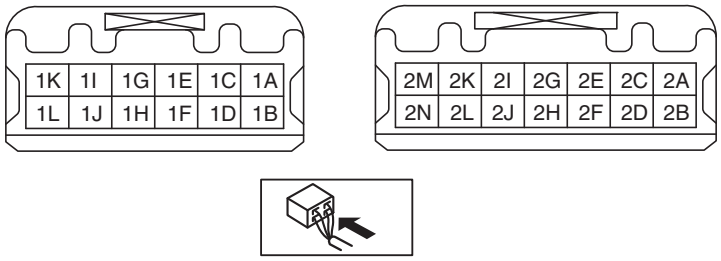
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Note
- The power outer mirror switch is built into the power window main switch.

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
2. Remove the power window main switch. (See POWER WINDOW MAIN SWITCH REMOVAL/INSTALLATION.)
3. Connect the power window main switch connector.
4. Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
5. Verify that the voltages of each of the terminals are as indicated in the terminal voltage table (reference).
   - If the voltage is not as specified in the terminal voltage table (reference), inspect the parts under Inspection item(s) and related wiring harnesses.
     — If the system does not work normally even though the inspection items are normal, replace the power window main switch.

Terminal Voltage Table (Reference)

POWER WINDOW MAIN SWITCH/POWER OUTER MIRROR SWITCH
WIRING HARNESS SIDE CONNECTOR



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Power Window Main Switch

L.H.D.

Terminal	Signal name	Connected to	Measurement condition	Voltage (V)	Inspection item(s)
1A	Open signal	Power window subswitch (RR)	Power cut switch is off (power cut switch is not pressed) and rear door glass (right side) is performing open operation	B+	• Power window motor (RR) • Power window subswitch (RR)
			Other	1.0 or less	
1B	Open signal	Power window subswitch (passenger's side)	Power cut switch is off (power cut switch is not pressed) and front door glass (passenger's side) is performing open operation	B+	• Power window motor (passenger's side) • Power window subswitch (passenger's side)
			Other	1.0 or less	
1C	Close signal	Power window subswitch (RR)	Power cut switch is off (power cut switch is not pressed) and rear door glass (right side) is performing close operation	B+	• Power window motor (RR) • Power window subswitch (RR)
			Other	1.0 or less	
1D	Close signal	Power window subswitch (passenger's side)	Power cut switch is off (power cut switch is not pressed) and front door glass (passenger's side) is performing close operation	B+	• Power window motor (passenger's side) • Power window subswitch (passenger's side)
			Other	1.0 or less	
1E	—	—	—	—	—
1F	Door open/close signal	BCM	Terminal used for communication therefore determination based on terminal voltage is not possible.		

Terminal	Signal name	Connected to	Measurement condition	Voltage (V)	Inspection item(s)
1G	Open signal	Power window subswitch (LR)	Power cut switch is off (power cut switch is not pressed) and rear door glass (left side) is performing open operation	B+	• Power window motor (LR) • Power window subswitch (LR)
			Other	1.0 or less	
1H	Open signal	Power window motor (driver's side)	Front door glass is performing open operation	B+	Power window motor (driver's side)
			Front door glass is performing open operation during approx. 40 s after ignition is switched from ON (engine off or on) to off	B+	
			Other	1.0 or less	
1I	Close signal	Power window subswitch (LR)	Power cut switch is off (power cut switch is not pressed) and rear door glass (left side) is performing close operation	B+	• Power window motor (LR) • Power window subswitch (LR)
			Other	1.0 or less	
1J	Close signal	Power window motor (driver's side)	Front door glass is performing close operation	B+	Power window motor (driver's side)
			Front door glass is performing close operation during approx. 40 s after ignition is switched from ON (engine off or on) to off	B+	
			Other	1.0 or less	
1K	IG1	P.WINDOW1 30 A fuse	Ignition is switched ON (engine off or on)	B+	• P.WINDOW1 30 A fuse • Battery
1L	Power supply	P.WINDOW2 25 A fuse	Under any condition	B+	• P.WINDOW2 25 A fuse • Battery
2A	Ground	Body ground	Under any condition	1.0 or less	Body ground
2B	Hall effect switch ground	Power window motor (driver's side)	Under any condition	1.0 or less	Power window motor (driver's side)
2D	Hall effect switch No.1	Power window motor (driver's side)	Front door glass (driver's side) is operating	Wave pattern (See Inspection Using an Oscilloscope (Reference))	Power window motor (driver's side)
2F	Hall effect switch No.2	Power window motor (driver's side)	Front door glass (driver's side) is performing open operation	Wave pattern (See Inspection Using an Oscilloscope (Reference))	Power window motor (driver's side)
2G	—	—	—	—	—
2M	ACC	MIRROR 7.5 A fuse	Ignition is switched ON (engine off or on) or ACC	B+	• MIRROR 7.5 A fuse • Battery
			Ignition is switched to OFF	1.0 or less	
2N	Hall effect switch power supply	Power window motor (driver's side)	Ignition is switched ON (engine off or on)	B+	Power window motor (driver's side)
			Ignition is switched off (LOCK)	1.0 or less	

R.H.D.

Terminal	Signal name	Connected to	Measurement condition	Voltage (V)	Inspection item(s)
1A	Open signal	Power window subswitch (RR)	Power cut switch is off (power cut switch is not pressed) and rear door glass (right side) is performing open operation	B+	• Power window motor (RR) • Power window subswitch (RR)
			Other	1.0 or less	

Terminal	Signal name	Connected to	Measurement condition	Voltage (V)	Inspection item(s)
1B	Door open/ close signal	BCM	Terminal used for communication therefore determination based on terminal voltage is not possible.		
1C	Close signal	Power window subswitch (RR)	Power cut switch is off (power cut switch is not pressed) and rear door glass (right side) is performing close operation	B+	• Power window motor (RR) • Power window subswitch (RR)
			Other	1.0 or less	
1D	Close signal	Power window subswitch (passenger’s side)	Power cut switch is off (power cut switch is not pressed) and front door glass (passenger's side) is performing close operation	B+	• Power window motor (passenger’s side) • Power window subswitch (passenger’s side)
			Other	1.0 or less	
1E	—	—	—	—	—
1F	Open signal	Power window motor (driver's side)	Front door glass is performing open operation	B+	Power window motor (driver's side)
			Front door glass is performing open operation during approx. 40 s after ignition is switched from ON (engine off or on) to off	B+	
			Other	1.0 or less	
1G	Open signal	Power window subswitch (LR)	Power cut switch is off (power cut switch is not pressed) and rear door glass (left side) is performing open operation	B+	• Power window motor (LR) • Power window subswitch (LR)
			Other	1.0 or less	
1H	Close signal	Power window motor (driver's side)	Front door glass is performing close operation	B+	Power window motor (driver's side)
			Front door glass is performing close operation during approx. 40 s after ignition is switched from ON (engine off or on) to off	B+	
			Other	1.0 or less	
1I	Close signal	Power window subswitch (LR)	Power cut switch is off (power cut switch is not pressed) and rear door glass (left side) is performing close operation	B+	• Power window motor (LR) • Power window subswitch (LR)
			Other	1.0 or less	
1J	Open signal	Power window subswitch (passenger’s side)	Power cut switch is off (power cut switch is not pressed) and front door glass (passenger's side) is performing open operation	B+	• Power window motor (passenger’s side) • Power window subswitch (passenger’s side)
			Other	1.0 or less	
1K	IG1	P.WINDOW1 30 A fuse	Ignition is switched ON (engine off or on)	B+	• P.WINDOW1 30 A fuse • Battery
1L	Power supply	P.WINDOW2 25 A fuse	Under any condition	B+	• P.WINDOW2 25 A fuse • Battery
2A	Ground	Body ground	Under any condition	1.0 or less	Body ground
2B	Hall effect switch ground	Power window motor (driver's side)	Under any condition	1.0 or less	Power window motor (driver's side)
2D	Hall effect switch No.1	Power window motor (driver's side)	Front door glass (driver's side) is operating	Wave pattern (See Inspection Using an Oscilloscope (Reference))	Power window motor (driver's side)

Terminal	Signal name	Connected to	Measurement condition	Voltage (V)	Inspection item(s)
2F	Hall effect switch No.2	Power window motor (driver's side)	Front door glass (driver's side) is performing open operation	Wave pattern (See Inspection Using an Oscilloscope (Reference))	Power window motor (driver's side)
2G	—	—	—	—	—
2M	ACC (Without door lock switch)	MIRROR 7.5 A fuse	Ignition is switched ON (engine off or on) or ACC	B+	• MIRROR 7.5 A fuse • Battery
			Ignition is switched to off	1.0 or less	
	Hall effect switch power supply (With door lock switch)	Power window motor (driver's side)	Ignition is switched ON (engine off or on)	B+	Power window motor (driver's side)
			Ignition is switched off (LOCK)	1.0 or less	
2N	Hall effect switch power supply (Without door lock switch)	Power window motor (driver's side)	Ignition is switched ON (engine off or on)	B+	Power window motor (driver's side)
			Ignition is switched off (LOCK)	1.0 or less	
	ACC (With door lock switch)	MIRROR 7.5 A fuse	Ignition is switched ON (engine off or on) or ACC	B+	• MIRROR 7.5 A fuse • Battery
			Ignition is switched to off	1.0 or less	

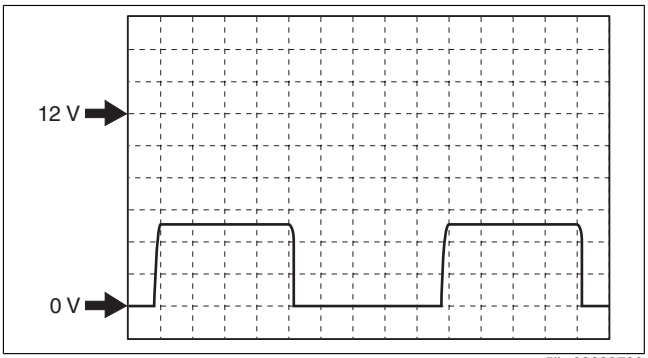
Power Outer Mirror Switch

Terminal	Signal name	Connected to	Measurement condition	Voltage (V)	Inspection item(s)
1L	Power supply	P.WINDOW2 25 A fuse	Under any condition	B+	• P.WINDOW2 25 A fuse • Battery
2A	Ground	Body ground	Under any condition	1.0 or less	Body ground
2C	Power outer mirror right/ down signal	Power outer mirror (LH)	Outer mirror glass (LH) is operating downward or to right direction	B+	Power outer mirror
			Other	1.0 or less	
2E	Power outer mirror retract signal	Power outer mirror (LH/RH)	Power outer mirror (LH/RH) are retracting	B+	Power outer mirror
			Other	1.0 or less	
2H	Power outer mirror right/ down signal	Power outer mirror (RH)	Outer mirror glass (RH) is operating downward or to right direction	B+	Power outer mirror
			Other	1.0 or less	
2I	Power outer mirror left signal	Power outer mirror (LH)	Outer mirror glass (LH) is operating to left direction	B+	Power outer mirror
			Other	1.0 or less	
2J	Power outer mirror up signal	Power outer mirror (LH/RH)	Outer mirror glass (LH/RH) is operating upward	B+	Power outer mirror
			Other	1.0 or less	
2K	Power outer mirror return signal	Power outer mirror (LH/RH)	Power outer mirror (LH/RH) are returning	B+	Power outer mirror
			Other	1.0 or less	
2L	Power outer mirror left signal	Power outer mirror (RH)	Outer mirror glass (RH) is operating to left direction	B+	Power outer mirror
			Other	1.0 or less	

Terminal	Signal name	Connected to	Measurement condition	Voltage (V)	Inspection item(s)
2M	ACC (Without door lock switch)	MIRROR 7.5 A fuse	Ignition is switched ON (engine off or on) or ACC	B+	• MIRROR 7.5 A fuse • Battery
			Ignition is switched to off	1.0 or less	
	Hall effect switch power supply (With door lock switch)	Power window motor (driver's side)	Ignition is switched ON (engine off or on)	B+	Power window motor (driver's side)
			Ignition is switched off (LOCK)	1.0 or less	
2N	ACC (With door lock switch)	MIRROR 7.5 A fuse	Ignition is switched ON (engine off or on) or ACC	B+	• MIRROR 7.5 A fuse • Battery
			Ignition is switched to off	1.0 or less	

Inspection Using an Oscilloscope (Reference)

- Connection terminal:
  2D (+) ↔ body ground (-)
  2F (+) ↔ body ground (-)
- Oscilloscope setting: 2 V/DIV (Y), 1 ms/DIV (X), DC range



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